# Menstrual tampons: HMTc Standard

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CORRESPONDENCE

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PUBLICATION HISTORY

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## Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for menstrual tampons (Feminine Care), evaluated on the total heavy-metal content of the finished article as placed on market (mucosal basis; the applicator is tested as a distinct component). No government sets a numeric heavy-metal content limit for this device form, so every analyte is held under mandatory analytical control or a substantiated not-material determination.

## 1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default). Device forms carry no sovereign heavy-metal content limit, so every panel cell resolves to a mandatory analytical control or a substantiated not-material determination; no occurrence percentile is computed and no number is invented. Cosmetic forms read across the strictest sovereign mucosal-contact cosmetic contaminant limit. Two basis families (device content, cosmetic content) are never conflated, and neither is a leachate/migration number. Provenance vocabulary: {regulatory-alignment, read-across, not-material, control, ratchet}.

## 2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 500 | human-grade read-across (DE) |
| Cadmium (Cd) | Tier 1 | 100 | human-grade read-across (DE) |

| | | | |
|---|---|---|---|
| Total arsenic (tAs) | Tier 2 | 500 | human-grade read-across (DE) |
| Inorganic arsenic (iAs) | Tier 1 | ≤ tAs | bounded by tAs (reflex speciation) |
| Mercury, total (tHg) | Tier 1 | 100 | human-grade read-across (DE) |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 10,000 | human-grade read-across (South Korea) |
| Tin (Sn) | Tier 2 | 3,750,000 | human-grade read-across (European Union) |
| Aluminium (Al) | Tier 2 | 560,000 | human-grade read-across (European Union) |
| Total chromium (Cr) | Tier 2 | 50 | human-grade read-across (AU) |
| Hexavalent chromium (Cr-VI) | Tier 2 | 5 | human-grade read-across (European Union) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 500 | DE — human-food analog |
| Cadmium (Cd) | 100 | DE — human-food analog |
| Total arsenic (tAs) | 500 | DE — human-food analog |
| Inorganic arsenic (iAs) | ≤ tAs | iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Mercury, total (tHg) | 100 | DE — human-food analog |

| | | |
|---|---|---|
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 10,000 | South Korea — human-food analog |
| Tin (Sn) | 3,750,000 | European Union — human-food analog |
| Aluminium (Al) | 560,000 | European Union — human-food analog |
| Total chromium (Cr) | 50 | AU — human-food analog |
| Hexavalent chromium (Cr-VI) | 5 | European Union — human-food analog |

## 4 | Cite this standard

Pendergrass, K. (2026). *Menstrual tampons: HMTc standard* (Version 1.0). Institute of Contaminant Standards.

## References


[1] AU — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] DE — human-food maximum levels
Binds Pb, Cd, tAs, tHg. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] European Union — human-food maximum levels
Binds Sn, Al, Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[4] South Korea — human-food maximum levels
Binds Ni. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL


### STATUS AND LICENCE


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### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

### DATA AVAILABILITY

### PUBLICATION HISTORY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/-menstrual-tampons. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-menstrual-tampons.